

Spectral Regions by Plant Trait

Key absorption features, mechanisms, and references for hyperspectral trait retrieval

Trait	Key Spectral Region(s)	Mechanism	Key References
Chlorophyll a+b	550-560 nm (green peak) 670-700 nm (red absorption) 700-750 nm (red edge)	Pigment absorption; red-edge shift with concentration changes	Feret et al. 2008 <i>PROSPECT-4/5</i> Feret et al. 2017 <i>Extended PROSPECT, Chl a/b separation</i> Blackburn 2007 <i>Review of pigment spectroscopy</i> Datt et al. 2015 <i>Meta-analysis of pigment detection</i>
Carotenoids	500-580 nm (blue-green)	Carotenoid absorption; overlaps with Chl in blue-green region	Feret et al. 2022 <i>PROSPECT-PMP+</i> Feret et al. 2008 <i>PROSPECT-4/5</i> Datt et al. 2015 <i>Meta-analysis</i>
Nitrogen	1510-1680 nm (SWIR-1) 2060-2150 nm (SWIR-2)	N-H stretching & combination bands; signal tied to protein and Rubisco content	Kokaly et al. 2013 <i>PNAS foliar N review</i> Zhang et al. 2020 <i>Advances in hyperspectral RS of vegetation traits</i> Serbin et al. 2012 <i>Leaf reflectance and photosynthetic metabolism</i>
LMA	1700-2400 nm (SWIR)	Dry matter and structural carbohydrate absorption; broad SWIR response	Asner et al. 2009 <i>Airborne spectranomics</i> Singh et al. 2015 <i>Imaging spectroscopy for canopy traits</i> Zhang et al. 2020 <i>Advances in hyperspectral RS of vegetation traits</i> Ely et al. 2025 <i>Full-spectrum hyperspectral modeling of leaf dry matter</i>

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Vcmax / Jmax	450-2500 nm (full spectrum; no single diagnostic band)	Captured via covariance with N, Rubisco, and Chl; requires PLSR or ML across full spectrum	Serbin et al. 2012 <i>Foundational PLSR for Vcmax/Jmax (JXB)</i> Ainsworth et al. 2019 <i>High-throughput phenotyping, PLSR (RSE)</i> Wu et al. 2024 <i>Deep learning for photosynthetic capacity</i> Magney et al. 2025 <i>Multi-trait hyperspectral integration (New Phytologist)</i>
Cellulose	2000-2100 nm 2280-2350 nm	C-H and O-H combination and overtone bands in SWIR-2	Serrano et al. 2015 <i>PROSPECT for protein, cellulose+lignin</i> Li et al. 2025 <i>SWIR imaging for cellulose, hemicellulose, lignin</i> Sanches et al. 2023 <i>Chemometrics UV-VIS-NIR-SWIR hyperspectroscopy</i>
Lignin	1680-1750 nm 2270-2350 nm	Aromatic C-H stretching; overlaps with cellulose in SWIR-2	Serrano et al. 2015 <i>PROSPECT for protein, cellulose+lignin</i> Li et al. 2025 <i>SWIR imaging for cellulose, hemicellulose, lignin</i> Sanches et al. 2023 <i>Chemometrics UV-VIS-NIR-SWIR hyperspectroscopy</i>
Water (EWT)	970 nm, 1200 nm 1450 nm, 1940 nm	O-H stretching overtones; strong signal must be accounted for when retrieving structural traits	Feret et al. 2008 <i>PROSPECT-4/5</i> Feret et al. 2017 <i>Extended PROSPECT</i>

Note: Spectral regions are approximate and may shift with species, leaf water content, and measurement conditions. EWT = Equivalent Water Thickness; LMA = Leaf Mass per Area; SWIR = Shortwave Infrared; PLSR = Partial Least Squares Regression.